

Predicting Type 2 Diabetes Risk

Final Report

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Introduction

As we know diabetes mellitus type 2 has become an increasingly prevalent disease in our population. According to the CDC, about every 1 in 10 Americans has diabetes. That amounts to 37 million Americans, of which 90 to 95% of cases can be attributed specifically to type II diabetes. It is therefore a subject worthy of investigation to understand as much as we can about the risk factors and interaction of those risk factors in developing DM 2. Understanding these risks will help to better shape how we can take preventative measures for those at risk of becoming diabetic and reduce the prevalence of diabetes.

Problem Definition

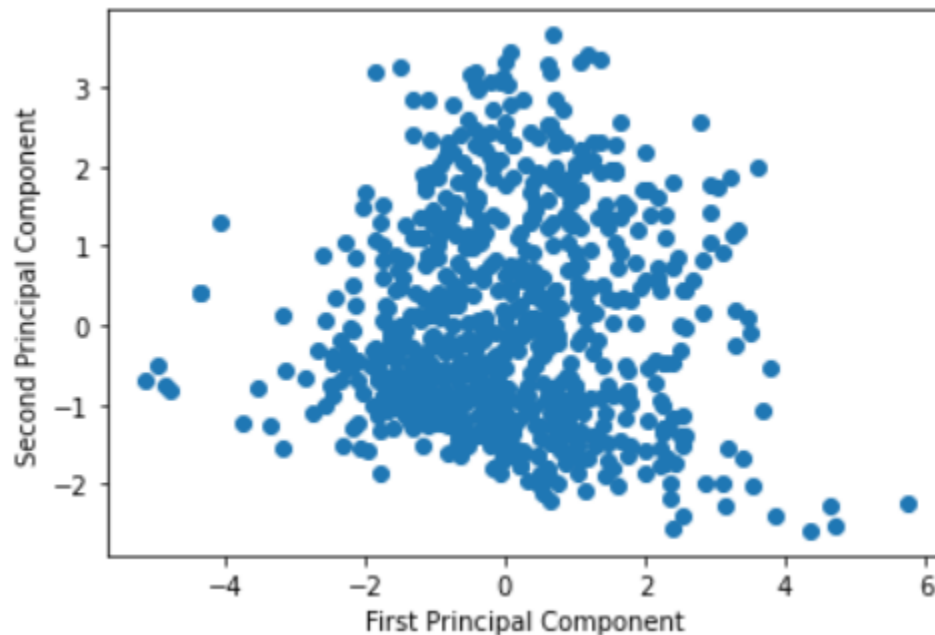
Our dataset on diabetes in Pima Indians has ~700 rows 9 attributes including the number of pregnancies, diastolic blood pressure, body mass index, diabetes pedigree, age, and outcome. Using this dataset we will create a risk stratification of factors or combinations of factors together that are more likely to create an increased risk of developing DM 2 compared to other factors or combinations of factors. Hence we will compare different groupings of factors relative to each other with respect to their associated risks of developing DM 2. Our investigation will seek to address this and better understand the interactions of the different factors with each other.

Methods

Methods that will be used include various clustering algorithms such as k-means clustering, hierarchical clustering, and density-based clustering. That way we can evaluate different groups with different combinations of risk factors and judge the effect of different combinations of risk factors on developing diabetes. Since we also have the labels for our data we will also seek to find the best predictive model for developing diabetes by using training, validation, and testing data sets and algorithms such as SVM and logistic regression to as well find the best predictive model.

Results and Discussion

Let's first visualize the data to get a general idea of how our data is distributed or centered. To do this, we will use PCA to reduce the data to 2 dimensions using the first 2 principal components.



No missing data was present, and no incorrect data types were present in the dataset. Data preprocessing included standardization for PCA and cluster analysis.

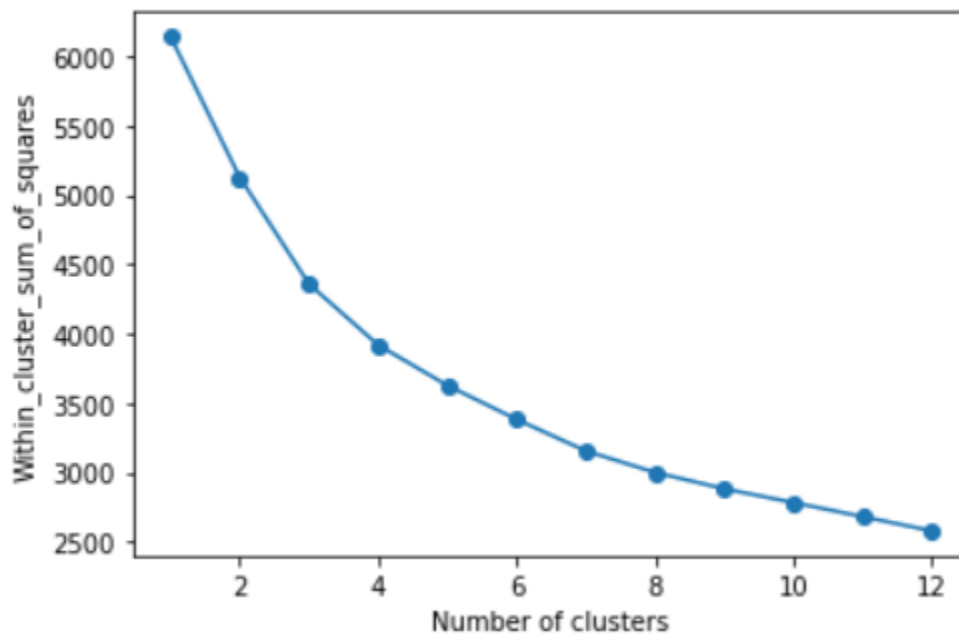
This plot seems to indicate that a lot of our data points fall in a contiguous range with a few outliers to the far left and far right. Because of the context of our analysis in dealing with diabetes, we are interested to see the effect of outliers and will keep them in the data. Outliers may provide information about the likelihood of developing diabetes and the interaction of different factors with each other and overall are data that could actually occur in a real scenario, so they will be kept to ensure a generalizable model.

K-Means

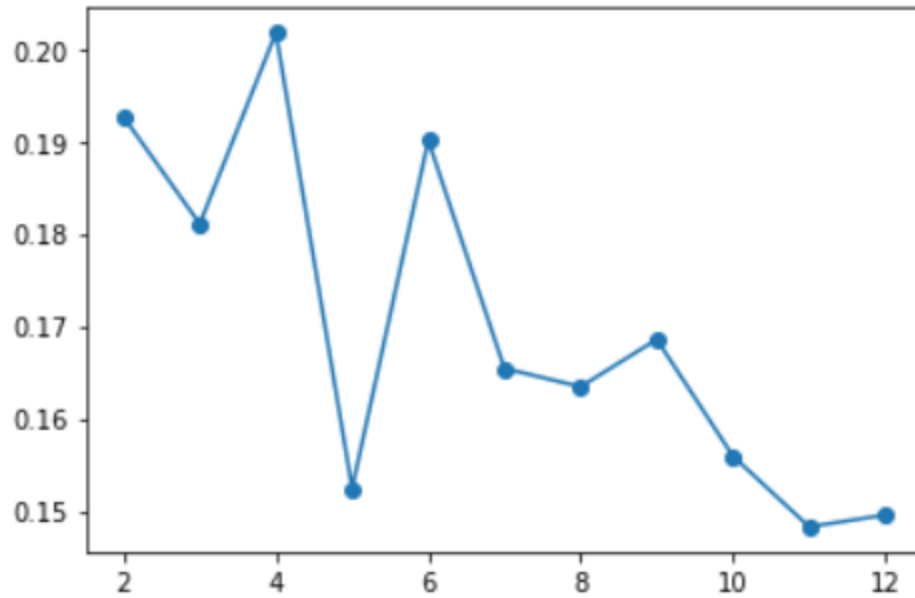
In this section of our investigation we will look into k-means clustering as a method of clustering our data into different groups. The purpose of clustering our data into different groups will be to look for any common factors or patterns among groups that could be significant with respect to the risk stratification of the Pima Indians. That way we could better understand which groups with certain characteristics may be more likely to develop diabetes.

Because of the objective of our clustering, we will use all of the features except the label to cluster into groups since we are interested in any potential differences in the clusters with respect to all the features.

To find the optimal number of clusters, we used an inertia score (which measures the sum of squared distances to the closest cluster for each data point) and a silhouette coefficient (which accounts for distances to a data point's cluster and the distance to its closest cluster it is not a member of).



The above plot shows inertia scores for a different number of clusters



The above plot shows silhouette coefficients for a different number of clusters

From what we can see from evaluating the different number of possible clusters, the optimal number of clusters appears to be 4 according to our evaluation metrics. This is because the inertia was lowest for 4 clusters and the silhouette coefficient was the highest for 4 clusters as seen in the plots above (the higher the silhouette coefficient, the more distinct the clusters). We can say with reasonable confidence that 4 is the optimal number of clusters for this data.

Taking a closer look at using 4 clusters, we get a homogeneity score of 0.15 indicating that with respect to our labels (whether the individual is diabetic or not) our four clusters have a mix of both diabetic and non-diabetic individuals. We also get a completeness score of 0.08 indicating that diabetic and non-diabetic patients are fairly dispersed among the clusters. Note the goal is not to have a high homogeneity or completeness score as the objective of the clustering is risk stratification rather than classifying an individual as diabetic or not.

After partitioning all of the data to their respective clusters we will now use a non-parametric test (Kruskal Wallis) to see if the marginal distributions for each of the features among different clusters are significantly different. A p-value of 0.05 was used for all of the Kruskal Wallis tests. This means that because we have 9 features and $(4C2)$ 6 different combinations of pairs of clusters, we will do a total of 36 comparisons.

	1,2	1,3	1,4	2,3	2,4	3,4
Pregnancies	True	False	False	True	True	True
Glucose	True	True	True	True	True	True
BloodPressure	True	True	True	True	True	True
SkinThickness	True	True	True	True	True	True
Insulin	True	True	True	True	True	True
BMI	True	False	True	True	True	True
DiabetesPedigreeFunction	False	False	True	False	True	True
Age	True	True	False	True	True	True
Outcome	False	True	False	True	True	True

The table above shows all of the comparisons. The columns show the pairs of clusters that were compared and the rows show which marginal distribution the comparison was done with. The first entry compares the marginal distributions of pregnancies for clusters 1 and 2. Any comparison where the null hypothesis was rejected such that the marginal distributions are significantly different is indicated with a true. The table below shows the medians with respect to each feature for each cluster.

	1	2	3	4
Pregnancies	2.000	7.0000	2.000	2.000
Glucose	116.500	126.0000	102.000	146.000
BloodPressure	0.000	78.0000	68.000	74.000
SkinThickness	0.000	0.0000	21.000	35.000
Insulin	0.000	0.0000	41.500	185.000
BMI	28.650	32.0000	29.500	36.500
DiabetesPedigreeFunction	0.292	0.3145	0.336	0.546
Age	28.000	45.0000	25.000	29.000
Outcome	0.000	0.0000	0.000	1.000

By combining the results of these 2 tables, we can obtain a form of risk stratification. The first table shows that clusters 1 and 2 have significantly different marginal distributions with respect to pregnancies and the second table shows that the median value of pregnancies is higher for cluster 2 than for cluster 1. Therefore, we can say that individuals from cluster 2 are more likely to have a higher number of pregnancies. On the other hand, we can not say that there is significant reason to believe that an individual from cluster 1 is more likely to be older or younger than an individual from cluster 4.

This would be useful for risk stratification because if we can predict to which cluster an individual belongs, we can see with respect to which features they are likely to differ or not differ from individuals in different clusters.

In an attempt to use k-means for classification purposes, we will use a total number of 2 clusters for the algorithm and evaluate its performance.

Confusion Matrix:

[[421 79]					
[182 86]]					
		precision	recall	f1-score	support
	0	0.70	0.84	0.76	500
	1	0.52	0.32	0.40	268
accuracy				0.66	768
macro avg		0.61	0.58	0.58	768
weighted avg		0.64	0.66	0.64	768

As we can see, using k-means for classification purposes with 2 clusters gives a precision score of 0.7 for classifying individuals as non-diabetic, and a precision score of 0.52 for classifying individuals as diabetic, suggesting that this model is probably better suited to classifying non-diabetic individuals.

DBscan

As part of the analysis of clustering algorithms, the DBscan algorithm was used to determine how the data is naturally clustered. The algorithm proved to be poorly suited for usage in this case - small interpoint distances resulted in large portions of the data being classified as noise, while larger distances ensured all data was part of a single cluster.

Values in the middle often had a small number of clusters, but nearly all of the data was part of the largest cluster formed. Smaller clusters occasionally showed some correlation with diabetes but had insufficient data to produce conclusions.

Hierarchical Clustering

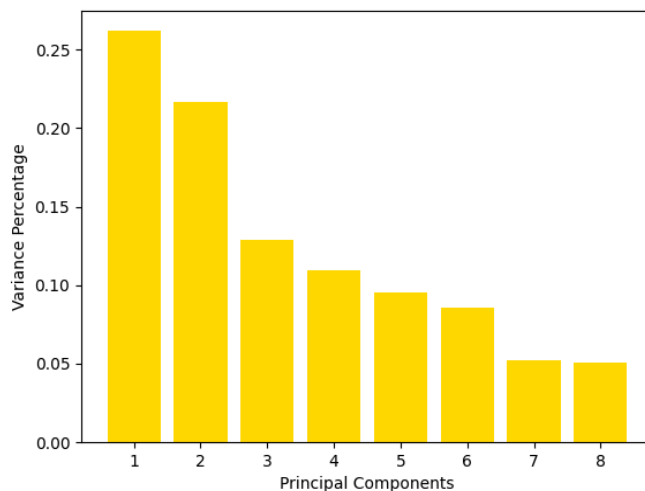
Due to the lack of data and 8 features, we checked the data correlation to see if any of them can be removed due to redundancy.

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age
Pregnancies	1	0.129458671	0.141281977	-0.081671774	-0.073534614	0.017683091	-0.033522673	0.544341228
Glucose	0.129458671	1	0.152589587	0.057327891	0.33135711	0.221071069	0.1373373	0.26351432
BloodPressure	0.141281977	0.152589587	1	0.207370538	0.088933378	0.281805289	0.041264948	0.239527946
SkinThickness	-0.081671774	0.057327891	0.207370538	1	0.43678257	0.392573204	0.183927573	-0.113970262
Insulin	-0.073534614	0.33135711	0.088933378	0.43678257	1	0.197859056	0.185070929	-0.042162955
BMI	0.017683091	0.221071069	0.281805289	0.392573204	0.197859056	1	0.140646953	0.03624187
DiabetesPedigreeFunction	-0.033522673	0.1373373	0.041264948	0.183927573	0.185070929	0.140646953	1	0.033561312
Age	0.544341228	0.26351432	0.239527946	-0.113970262	-0.042162955	0.03624187	0.033561312	1

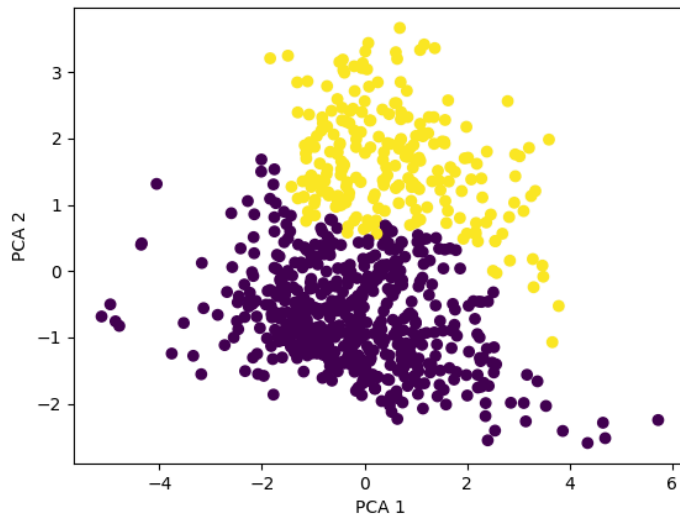
From the correlation table, there is no significant correlation except for the highest correlation of 0.544 between Age and Pregnancy number, so the approach is not good enough and there is nothing we can do here.

Using the Principal Component Analysis (PCA) algorithm is a good approach to linearly reduce dimensions. Before using PCA, we normalized our data between features with the formula $\frac{value - mean}{standard deviation}$ because PCA relies on the variance of each feature.

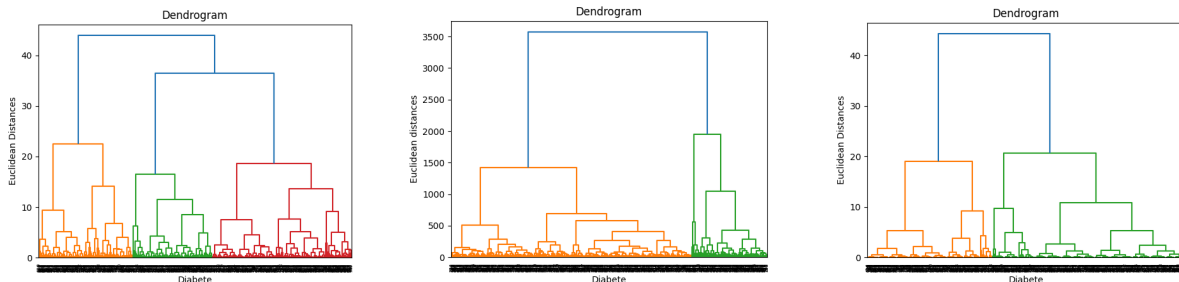
After using PCA, we plotted the variances for each principal component and figured out that the first and second principal components have the most variance so we used them for training with the agglomerative clustering.



When we plot PC 1 and 2 for visualization, we can kind of see 2 clusters as we expect.



To find the optimal number of clusters, we generated dendrograms using the original dataset, PC 1 and 2, PC 1 respectively.

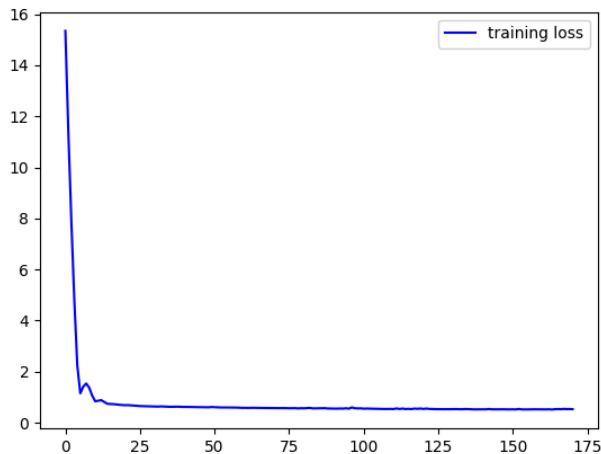


Although these dendrograms all show that the optimal number of clusters should be 4, we decided to train the agglomerative clustering with the assumption of 2 clusters due to the binary outcome of diabetes. From the original data input to data through PCA, the clustering result has improved significantly.

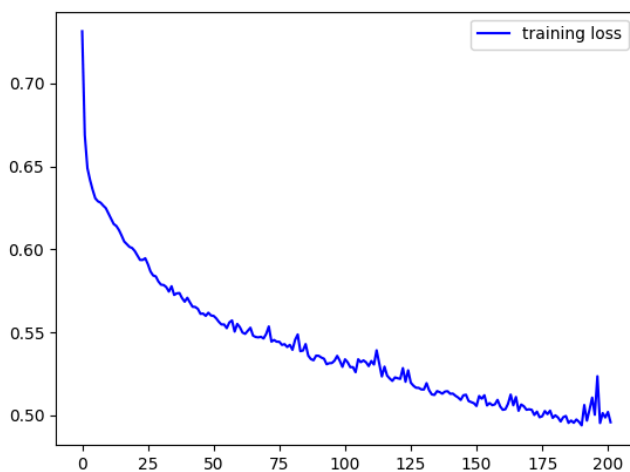
	Before PCA	Using PC 1 and 2	Using only PC 1
Precision Score	0.285	0.550	0.564
Recall Score	0.623	0.474	0.638
F1 Score	0.391	0.509	0.599
Accuracy Score	0.324	0.681	0.702

Neural Network

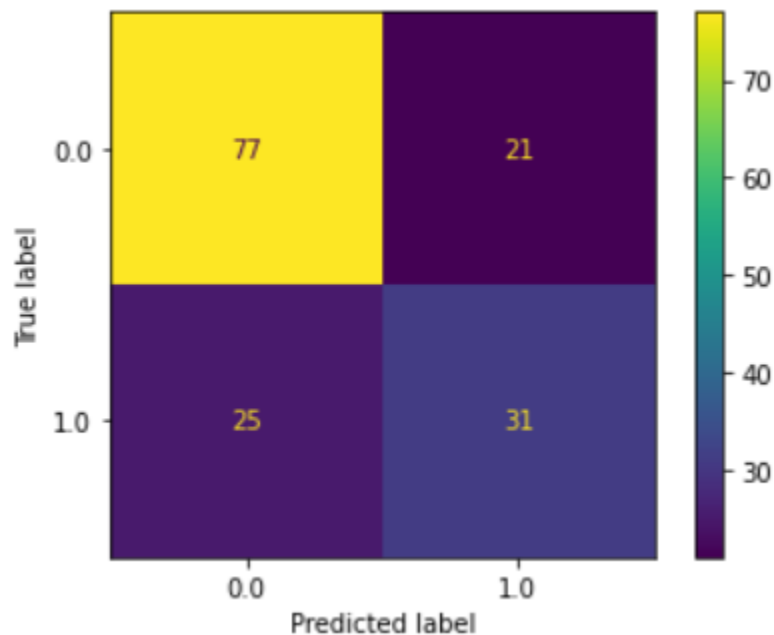
Neural network is used to discover existing patterns in data and extrapolate them. Thus, it is better for classification than prediction. At the first tryout, neural network already returned a pretty good accuracy result compared to other methods' first tryout. The training loss graph looks quite absurd but the neural network outperformed other methods very well in terms of minimizing the loss.



After using a grid search to test the neural network with multiple different combinations of parameters, we found the possible best parameters for activation function, alpha, learning rate, etc which gave us the possible best prediction model. The training loss graph looks more natural than the first tryout. The result is a little bit better but not significantly improved. Neural network is harder to debug because it is kind of a black box and the results are sometimes too inconsistent when I tried K-fold cross-validation compared to other methods.



The following shows a confusion matrix of the neural network model used on our testing data set.

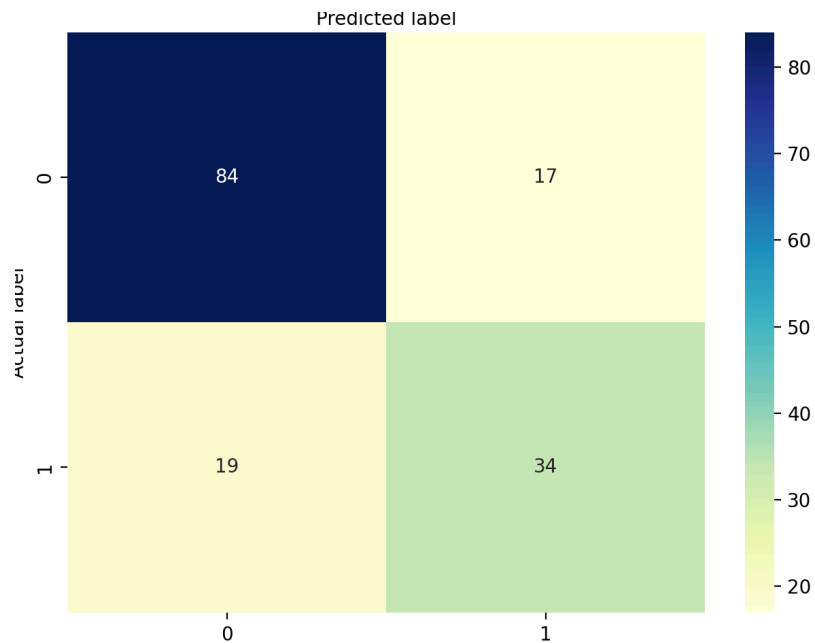


	Before GridSearch	After GridSearch
Precision Score	0.607	0.627
Recall Score	0.642	0.698
F1 Score	0.623	0.661
Accuracy Score	0.733	0.753

Logistic Regression

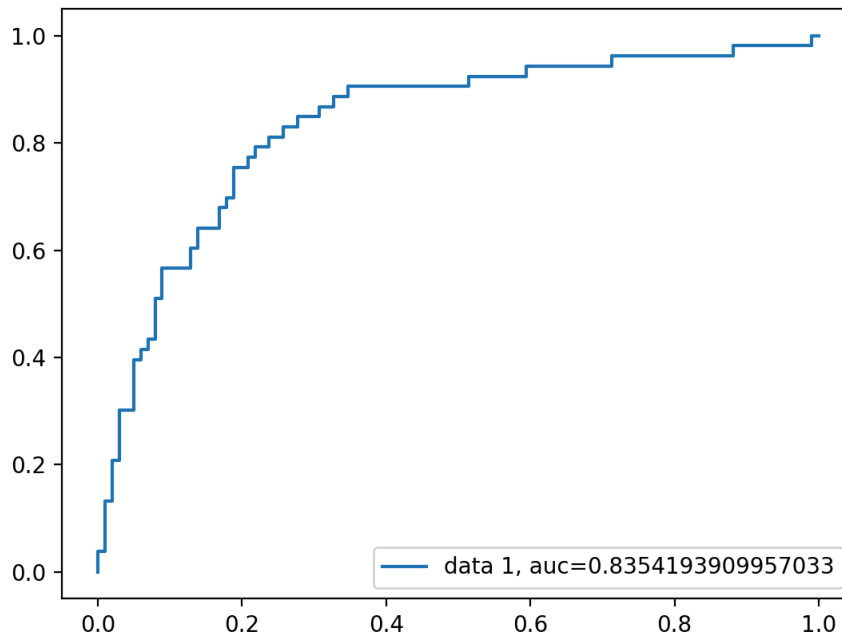
Logistic regression is used for modeling the probability of a discrete result given some input variables. After fitting the diabetes data into the model, we got a score of 0.7662 which represents the mean accuracy of the given test data and labels. Also, the recall score was 0.6415, the precision score was 0.6666, the F1 score was 0.6538, and the accuracy score was 0.7662.

The confusion matrix was $\begin{bmatrix} 84 & 17 \\ 19 & 34 \end{bmatrix}$. The dimension of this matrix is 2 by 2 since its a binary classification, a value of 1 represents the presence of diabetes and a value of 0 represents the absence of it. The diagonal values are accurate predictions (84 and 34), while non-diagonal elements are incorrect predictions (17 and 19).



The Receiver Operating Characteristic (ROC) curve is a plot of the true positive against the false positive rate. This curve is used to determine the cutoff value for predicting the output (0 or 1) of a new observation. This curve displays the sensitivity and specificity of a logistic regression model.

The area under the curve (AUC) for this example was 0.8354, where a score of 1 is a perfect classifier. So, we can conclude that this model does a good job of classifying the data.



SVM

Another classifier that was used on the data was the support vector machine classifier. To find the best SVM classifier we tested different kernels which include the linear, polynomial, radial basis function (RBF), and sigmoid kernels.

By splitting our data set into training, validation, and testing sets, we used cross-validation to find the SVM classifier that performed the best. One thing that should be considered is how we will evaluate which of these models performed best as there are several potential evaluation metrics that can be used such as accuracy, precision, and several others. Due to the context of the data we are dealing with, the best evaluation metric would most likely be the area under the curve of the ROC curve. This is because although metrics such as accuracy can give us an overall measure of how many data points were correctly classified, a very important aspect in the context of medical data is how many false negatives we had. To illustrate this, we consider a false negative and a false positive in the context of a diabetes mellitus type 2 diagnosis. It is of much higher priority to minimize false negatives than it is to minimize false positives because missing a diabetes diagnosis has far more implications than a false positive diabetes diagnosis due to the long term effects that untreated diabetes can have such as chronic kidney disease and emergency cases of severe spikes in glycemic levels. This is why the area under the curve of the ROC curve was used

to compare these models as it gives us an indication of how well this classifier does in minimizing false negatives and maximizing true positives.

The following table shows 5 scores corresponding to the area under the curves for the ROC plots for each classifier. Note 5 scores for each classifier since we did 5-fold cross-validation.

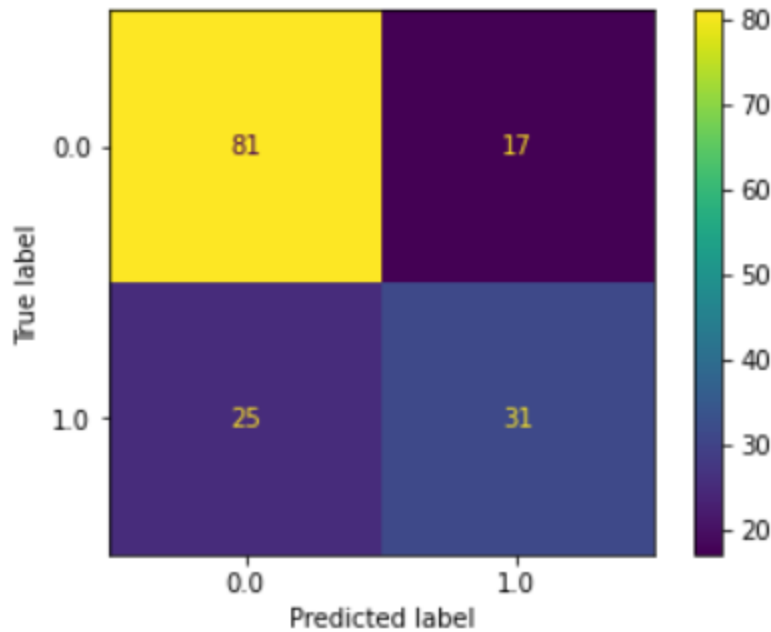
	scores_svmlinear	scores_svmpoly	scores_svmrbf	svmsigmoid
0	0.763081	0.753779	0.721221	0.293895
1	0.879360	0.834884	0.852035	0.227326
2	0.877131	0.858319	0.857731	0.228101
3	0.842740	0.825691	0.821282	0.315403
4	0.830060	0.825298	0.815774	0.318155

The following table shows the average of the scores for each classifier.

```
scores_svmlinear    0.838474
scores_svmpoly      0.819594
scores_svmrbf       0.813608
svmsigmoid          0.276576
dtype: float64
```

According to the average scores for each classifier, the best SVM classifier was the one that used the linear kernel.

The following confusion matrix shows the results of using the SVM classifier with the linear kernel on the test dataset. Note the AUC score for this specific model was 0.69.



Conclusion

In conclusion, we can divide our investigation into three main parts. The first part involved using clustering as a form of segmenting our population of interest into different groups for the purpose of risk stratification. This way one can see with respect to which features individuals from different groups are likely to differ. This means that this information can be combined with information from the previous literature to achieve a risk stratification. So if the literature has shown that feature X is more correlated with the development of DM type 2, we can use our clustering to predict which clusters are likely to have a significantly higher or lower value of feature X and hence which clusters may be more at risk for the development of DM type 2 with respect to that feature. The table showing the comparisons of different features among different clusters can be found in the K-means section. Note that in this part of the investigation 4 clusters were the optimal number of clusters for the risk stratification. We did not use 2 clusters as we did for the classification part, because we are not classifying but rather doing a risk stratification.

We then proceeded to build a classification model that can predict individuals as either having or not having DM type 2. We divided the models into supervised and unsupervised models depending on if the outcome label from the data was used. From the unsupervised models, the hierarchical clustering outperformed the k-means and the DBSCAN clusterings with respect to how accurate the models were. The first model of hierarchical clustering performed very poorly so the dimensionality of the data was reduced for that model and the DBSCAN was judged not to be a suitable model due to most of the data being classified

as either noise or as part of a single cluster depending on the epsilon parameter. Overall, classifying without the outcome of the datapoint hierarchical clustering would be the best performing.

From the supervised models of the neural network, the logistic regression, and the support vector machine the logistic regression performed the best. We can see this by looking at the confusion matrices and the number of false negatives and false positives with respect to each model. As discussed previously, because of the context of the data we would give a higher priority to a model that minimizes false negatives since it would have more implications to miss a DM type 2 diagnosis than to falsely diagnose DM type 2. This is why the AUC score and the number of false negatives of each of these models were important in choosing the best-performing model. The logistic regression model had both the highest AUC score and lowest classifications of false negatives which is why it was the best supervised model for predicting DM type 2.

References

- [1] Ferrannini E, Cushman WC. Diabetes and hypertension: the bad companions. Lancet. 2012 Aug 11;380(9841):601-10.
- [2] Fletcher B, Gulanick M, Lamendola C. Risk factors for type 2 diabetes mellitus. J Cardiovasc Nurs. 2002 Jan;16(2):17-23.
- [3] Zhu Y, Zhang C. Prevalence of Gestational Diabetes and Risk of Progression to Type 2 Diabetes: a Global Perspective. Curr Diab Rep. 2016 Jan;16(1):7.

Link to Github page: https://github.gatech.edu/pages/jtepper7/diabetes_ML/